

EXAM 2 PRACTICE PROBLEMS

1. Return to the email dataset from Lecture 7. This time, let's examine the variable "line_breaks", including the following steps:
 - a. Plot the variable using an appropriate R function.
 - b. Identify the sample mean, sample size, and sample standard deviation.
 - c. Obtain a 95% two-sided confidence interval for this variable.
 - d. Perform a hypothesis test at confidence level $\alpha=0.05$ to investigate whether or not the population mean for line breaks is 250. List both the test statistic and its resulting p-value.
 - e. Do you reject or fail to reject the null hypothesis?
 - f. What can you conclude about the mean number of line breaks given the above hypothesis test?

NOTE: You can read in the data using this code:

```
email <-
```

```
read.csv("https://raw.githubusercontent.com/roualdes/data/master/email.csv")
```

2. The average monthly energy use for a typical household was previously determined to be 909 kWh, with a known population standard deviation of 105 kWh. Say a survey is done to test whether the average energy use is now **higher** than this. Download the "energy.csv" file to view the results of the survey, which list the energy use in kWh of several homes.
 - a. Plot the variable using an appropriate R function. Do you notice any unusual values or outliers in the data? Why might such a value exist for this data set?
 - b. Identify the sample mean, sample size, and sample standard deviation for this variable.
 - c. Use the population standard deviation to find a 95% confidence interval for mean energy usage.
 - d. Perform an appropriate hypothesis test at confidence level $\alpha=0.05$ to investigate the above suspicion for this data. List both the test statistic and its resulting p-value.
 - e. Do you reject or fail to reject the null hypothesis?
 - f. What if you instead decide to use the sample standard deviation to perform a t-test? Find both the confidence interval and the hypothesis test for this situation.
 - g. Do your results differ from those of part e? Why or why not?

3. A medical study recorded the cholesterol levels of 25 patients in both 2010 and 2020, and wishes to determine whether there is a difference in cholesterol levels for each patient. Download the data set “cholesterol.csv” for this question.
- Would this data call for a paired t-test or a two-sample t-test?
 - Plot the variable using an appropriate R function.
 - Identify all relevant quantities (such as the estimator, sample sizes, and sample standard deviations) for this variable.
 - Obtain a 98% two-sided confidence interval for this variable.
 - Perform an appropriate hypothesis test at confidence level $\alpha=0.02$. List both the test statistic and its resulting p-value.
 - Do you reject or fail to reject the null hypothesis?
 - What can you conclude about the patients' cholesterol levels based on your results in part f?
4. In RStudio, generate a data set of size $n=20$ from the Exponential distribution, with $X \sim \text{Exp}(\text{rate}=5)$.
- Create a histogram for this sample. What skew, if any, does the data have?
 - Calculate $R=500$ sample statistics $\bar{X}$ for $r=1, \dots, R$.
 - Create a histogram for your sample statistics. How does the sampling distribution for the estimator $\bar{X}$ compare to that of the initial sample?
 - Repeat the above steps for a sample (and multiple samples) of size $n=200$. How do the population and sampling distributions change?